

From Translation to Grounded Dialogue Generation: Report

Ishaan (GitHub: vector298), WEC Intelligence SIG recruitment tasks 2026. Three LSTM-based sequence-to-sequence systems built from scratch in PyTorch (*nn.LSTM*, *nn.Embedding*, standard training loops): an English to French encoder-decoder, the same model with attention and a comparison of decoding strategies, and a document-grounded Hinglish dialogue generator. No pretrained transformers and no packaged seq2seq pipelines were used. Every number below is from the notebooks in *subtask1*, *subtask2* and *subtask3*; test sets were run once.

1. Approach and datasets

Subtasks 1 and 2: English to French translation

Data. Provided TED-style CSVs: train 232,825 pairs, validation 890, test 8,597. The corpus is noisy (misaligned pairs, such as a French line of "-" for an English sentence; stage directions like --Rires--). **Cleaning** (all splits): whitespace and apostrophe normalisation and removal of French stage directions by a whitelist (14 training segments). A first, looser rule that deleted any text between two dashes also removed real words from English em-dash asides; I found this by checking what the rule removed and replaced it. **Training filter only:** both sides 1 to 50 tokens and length ratio within 1/3 to 3 (220,442 of 232,825 pairs kept); validation and test are never filtered. Leakage: 12 of 890 validation and 96 of 8,597 test pairs also occur in train (1.6% of test English sentences); test BLEU is reported with and without them. **Tokenisation:** word level; splitting French after apostrophes (l' idee) lowered the validation <unk> rate from 7.06% to 5.88%, so it was chosen. Vocabularies of 10,000 words per language.

Models. Baseline: a 2-layer LSTM encoder-decoder (embedding 256, hidden 512, dropout 0.3; 17,606,416 parameters) whose decoder starts from the encoder's final state (one fixed context vector), with padding handled by packing and masking. Attention model: the same network plus Luong "general" attention (padding masked before the softmax; 18,393,360 parameters). **Protocol for both:** Adam 1e-3, batch 128 (length-bucketed), gradient clipping 1.0, scheduled teacher forcing 1.0 down to 0.5, 10 epochs, best checkpoint by greedy validation BLEU, seed 42, one Tesla T4, so any difference is due to attention. Metric: sacrebleu (lowercase, 13a) with 95% bootstrap intervals from per-sentence statistics. **Decoding** (compared on the 890 validation sentences): greedy; beam search (widths 3 and 5, length-normalised); temperature sampling (0.7, 1.0); top-k (10); top-p (0.9); sampled rows are means of 3 seeds.

Subtask 3: document-grounded Hinglish dialogue

Data. CMU Hinglish DoG: 8,060 / 942 / 960 utterances in 205 / 27 / 27 conversations (train / validation / test), giving 7,855 / 915 / 933 examples. The Hugging Face rows hold the utterances but not the documents; the 30 movie documents (4 sections each) come from the original English repository, and a turn's *docidx* selects the section being shown. **Example:** input = previous 3 turns (Hinglish, at most 60 tokens) plus the current document section (first 150 tokens); target = the Hinglish reply (first 30 tokens). One shared word-level vocabulary (9,630 words; <unk> 4.4% in validation replies and 9.0% in documents), with embeddings learned from scratch. **Grounded model:** two LSTM encoders (history, document; shared embedding), decoder state initialised from both, separate attention over each at every step (7,109,790 parameters). **Baselines:** the same model without the document (6,189,726 parameters; identical data, hyperparameters, seed and early stopping) and a TF-IDF retrieval baseline (return the reply of the most similar training history). Adam 1e-3, batch 128, early stopping on validation loss (at most 80 epochs; a first 12-epoch run was underfitted and replaced before the test set was used), greedy decoding. **Metrics:** BLEU, ROUGE-L, distinct-1/2, reply length and a grounding rate (the share of a reply's content words, those outside the 200 most frequent training words, that occur in the true document), plus a **document-swap test** (run the grounded model with documents shuffled across test examples).

2. Experimental results

Subtask 1 and 2: baseline against attention (test, one run)

System	Best val BLEU (epoch)	Test BLEU [95% interval]	BLEU, test rows not in train	Repeated-trigram outputs
No attention (greedy)	6.37 (9)	9.41 [9.17, 9.66]	9.37	22.8% (references 3.8%)
Attention (greedy)	18.18 (9)	25.50 [25.10, 25.87]	25.47	8.6%
Attention (beam 5)	n/a	27.72 [27.33, 28.11]	27.68	7.0%

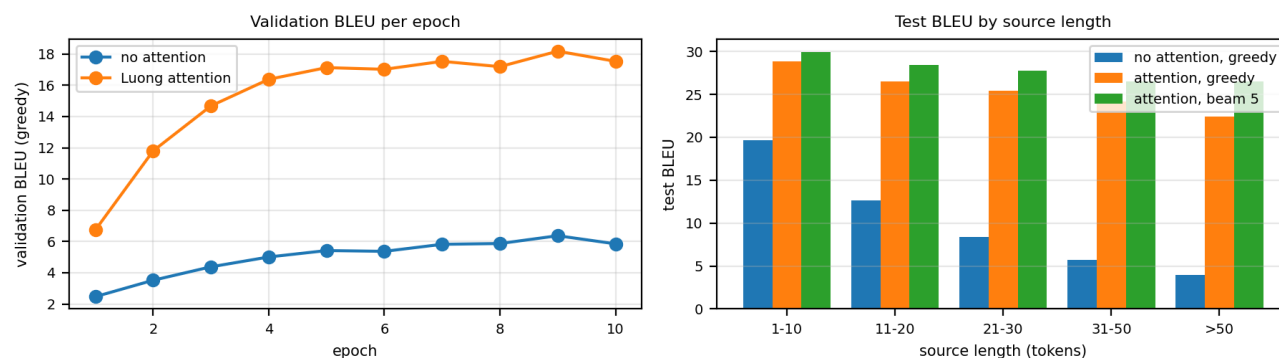


Figure 1 (plotted from the reported numbers). Left: validation BLEU per epoch at an equal 10-epoch budget. Right: test BLEU by source length.

At an equal budget attention is 2.9 times the baseline on validation BLEU (18.18 against 6.37) and already exceeds the baseline's best after one epoch (6.74). Validation loss ends at 2.154 against 2.885; an epoch takes about 251 s against 218 s.

The baseline was still improving at epoch 10 while attention plateaued near epoch 9, so a longer baseline run might narrow the gap (not measured). **By source length** the baseline falls from 19.66 BLEU (1 to 10 tokens, 2,325 test sentences) to 3.99 (over 50 tokens, 247), a drop of 15.7 points; attention falls from 28.85 to 22.42 (greedy, 6.4 points) and from 29.97 to 26.50 (beam 5, 3.5 points), so its gain grows with length (+9.2 points at 1 to 10 tokens, +18.4 over 50, greedy).

Failure modes of the baseline and the remaining weakness of attention

Unknown words: 43.6% of test sources contain an <unk>; baseline BLEU is 6.46 with and 13.96 without; attention 20.75 with and 32.26 without. 62.8% of baseline outputs contain <unk>. **Degeneration:** baseline outputs start correctly and then loop ("et, et, et", "ce que ce que"); 22.8% repeat a trigram, against 8.6% for attention and 3.8% for the references. Attention maps (two validation sentences) follow a rough diagonal and align content words (riviere with river, groenland with greenland, charbon with coal, serieux with serious) with local reordering; they also show two remaining problems: two generated "groenland" tokens attend to the same source word (no coverage mechanism) and a generated <unk> attends to the source <unk>, so the rare word is lost at the input. BLEU also penalises valid paraphrases ("et c'est ce qu'ils ont fait" for "Et elles se sont tues.").

Decoding strategies (attention model, validation, 890 sentences)

Strategy	BLEU	distinct-1	distinct-2	Repeated trigram %	Length ratio
greedy	18.18	0.191	0.604	10.6	0.969
beam 3	19.89	0.196	0.617	7.2	0.978
beam 5	20.21	0.197	0.622	6.9	0.979
temperature 0.7	13.61	0.209	0.663	5.2	0.979
temperature 1.0	8.13	0.250	0.763	1.5	0.999
top-k 10	10.55	0.207	0.667	5.1	0.989
top-p 0.9	10.31	0.228	0.719	2.7	0.991

Beam search adds 1.70 (width 3) and 2.03 (width 5) BLEU over greedy and cuts repetition from 10.6% to 6.9%; the step from width 3 to 5 adds only 0.32. Sampling lowers BLEU by 4.6 to 10 points but raises diversity (distinct-2 from 0.604 to 0.663 to 0.763) and lowers repetition; with one reference per sentence BLEU favours the single best output. Temperature 0.7 keeps the most BLEU among the sampling methods; I had expected top-p and top-k to beat it, and they did not. Greedy decoding is batched and beam search runs per sentence, so the time ratio (0.4 s against 24 s per 890 sentences) partly reflects the implementation.

Subtask 3: grounded against ungrounded against retrieval (test, 933 examples from 27 conversations)

System	Best val loss (epochs run)	BLEU	ROUGE-L	distinct-1 / 2	Length	Grounding
Retrieval (TF-IDF)	n/a	0.42	0.064	0.159 / 0.447	10.6	0.040
Ungrounded	5.114 (22)	0.72	0.108	0.014 / 0.055	6.7	0.000
Grounded	5.161 (23)	0.98	0.107	0.017 / 0.072	8.2	0.282
Human reference	n/a	n/a	n/a	0.208 / 0.684	12.6	0.100

The grounded model uses the document, but rarely. In the swap test (94% of examples received another movie's document) 91.7% of its outputs changed, its grounding to the true document fell from 0.282 to 0.047 and grounding to the swapped document rose to 0.391; ROUGE-L stayed 0.107 to 0.105 and BLEU 0.98 to 0.82. But only 10.1% of its replies contain any content word (ungrounded 2.8%, references 92.7%), so the grounding evidence rests on about 94 replies; 12.5% of its content words occur in the document (references 10.8%). It mostly copies names and titles (for example "la land"). **Both generators collapse into generic replies:** distinct-2 is 0.055 and 0.072 against 0.684 for people, replies are short, and the most common outputs are "kya" (43 and 44 times) and frames such as "mujhe lagta hai ki hai ."; the ungrounded model produced only 360 distinct replies out of 933, the grounded 485. Both overfit after about epoch 17 (train loss 4.68 and 4.95 against validation loss 5.16 and 5.11 at the best epoch; the grounded model, with more capacity, overfits more). Retrieval returns real replies but often for the wrong context (a Frozen reply in a La La Land conversation) and has the lowest overlap (ROUGE-L 0.064). By code-mixing proxy (terciles of the share of Hinglish tokens also present in the English version) ROUGE-L differs by only about 0.01 between the most Hindi-heavy and the most English-heavy third (0.106 to 0.110 against 0.098 to 0.099), within noise.

3. Insights and takeaways

- **Attention is the largest single improvement, and the evidence points at the fixed context vector.** Same data, budget and protocol: 2.9 times the validation BLEU, and a length curve that is about 2.5 times (greedy) to 4.5 times (beam 5) flatter. It is association measured on identical sentences, not a controlled proof, and the longest bucket (247 sentences) is beyond the training lengths.
- **Decoding is a quality-diversity trade-off, and the right choice depends on the task.** Beam search is better for translation (one right answer); sampling lowers BLEU but removes repetition, which is what open-ended dialogue needs.
- **A word-level vocabulary is now the binding limit in translation.** Sentences containing an unknown word still lose about 11.5 BLEU points after attention (20.75 against 32.26); a subword vocabulary or a copy mechanism is the obvious next step.
- **In dialogue, conditioning on a document changed what the model says without making it say more or better.** Measurable dependence on the document (the swap test) did not translate into higher ROUGE-L, lower validation loss or more varied replies. With about 7,900 training examples, maximum-likelihood training and greedy decoding, both models fall back on frequent words; the data size, not the architecture, is the probable bottleneck (not tested).

- **Evaluation choices matter as much as models.** BLEU is below 1 for every dialogue system, so ROUGE-L, diversity, grounding and the swap test carried the analysis; bootstrap intervals, equal budgets, a test set run once and a leakage check kept the comparisons honest. Inspecting what a cleaning rule actually removed caught a mistake that summary statistics would not have shown.

4. Limitations

One seed per model; the baseline was still improving at the end of its budget; validation BLEU (6.37 and 18.18) is lower than test BLEU (9.41 and 25.50) for both models, which I did not investigate; the decoding comparison uses the small validation set; no coverage mechanism, subword vocabulary or human evaluation; the subtask 3 test set has only 27 conversations, so every dialogue number is noisy, and the grounding rate and the code-mixing proxy are my own rough measures; the model sees only the current document section. **Setup:** Kaggle, Tesla T4, Python 3.13.15, torch 2.11.0+cu128, sacrebleu 2.6.0, seed 42; scikit-learn TF-IDF was used only for the retrieval baseline and Hugging Face *datasets* only to load the Hinglish data.